THE QUOT FUNCTOR IS AN FPQC SHEAF

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-quot (canonical tag 09TR). cited at statement line 2463. Standing situation.
- Lemma quot-lemma-q-sheaf (canonical tag 082P). cited at proof line 2468. Q^fp is a sheaf (collected card S7-q-q-sheaf).
- Lemma spaces-descent-lemma-descending-property-proper (canonical tag 0422). cited at proof line 2474. Descending property: proper.

Source reference. Original definition 09TR: sources/stacks-chow/tags.txt, source lines 12399–12399. Cited at statement line 2463: the standing situation of the Quot functor section.

Source reference. Original lemma 082P: sources/stacks-chow/tags.txt, source lines 10259–10259. Cited at statement line 2468: Q^fp is a sheaf.

Source reference. Original lemma 0422: sources/stacks-chow/tags.txt, source lines 5313–5313. Cited at statement line 2474: descending property: proper support.

Source reference. Original lemma 089C: sources/stacks-chow/tags.txt, source lines 10492–10492. Cited at statement line 2476: scheme-theoretic support commutes with flat base change.

Lemma 0.2. *In Situation 09TR. The functor* $\mathrm{Quot}_{\mathcal{F}/X/B}$ *satisfies the sheaf property for the fpqc topology.*

Proof. In Lemma 082P we have seen that the functor $\mathrm{Q}^{fp}_{\mathcal{F}/X/S}$ is a sheaf. Recall that for a scheme T over S the subset $\mathrm{Quot}_{\mathcal{F}/X/S}(T) \subset \mathrm{Q}_{\mathcal{F}/X/S}(T)$ picks out those quotients whose support is proper over T. This defines a subsheaf by the result of Descent on Spaces, Lemma 0422 combined with Morphisms of Spaces, Lemma 089C which shows that taking scheme theoretic support commutes with flat base change. □

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